

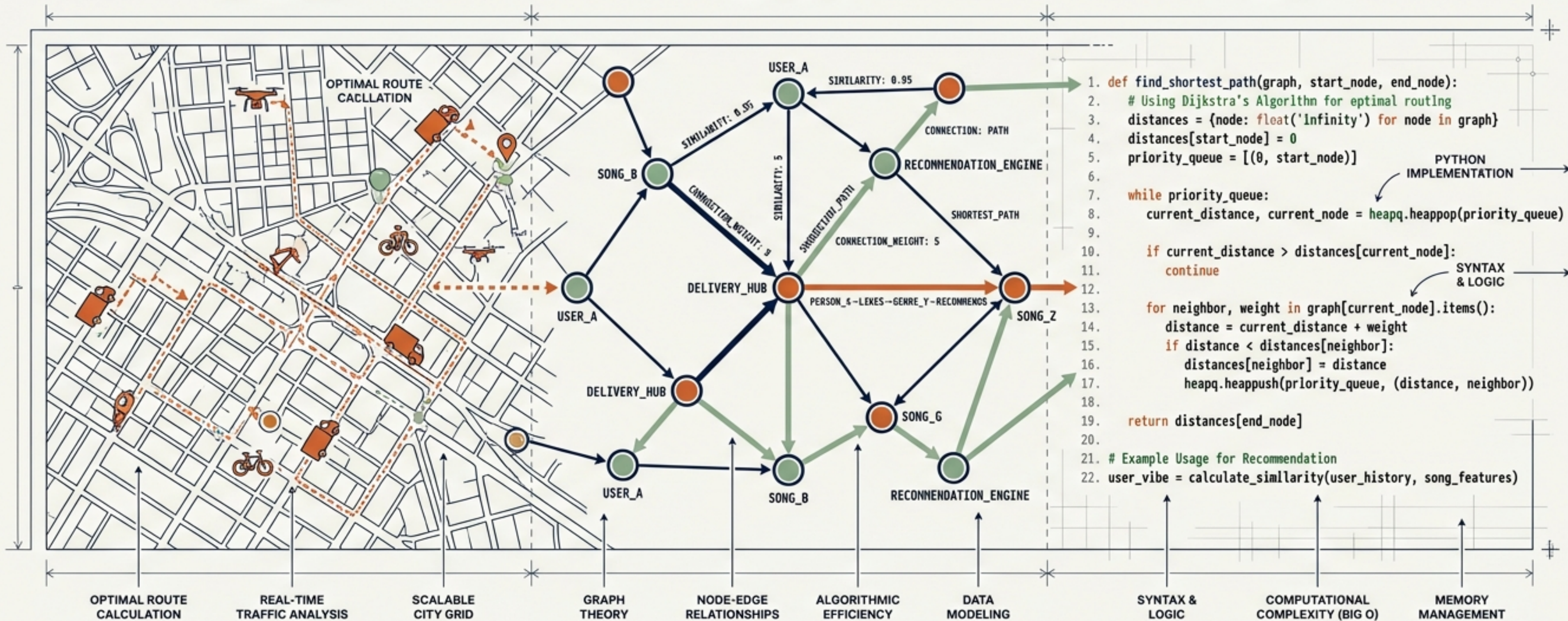
Data Structures & Algorithms: From Scale to Syntax

Why your delivery app crashes, how Spotify knows your vibe, and the code that makes it possible.

THE MACRO

THE MESO

THE MICRO

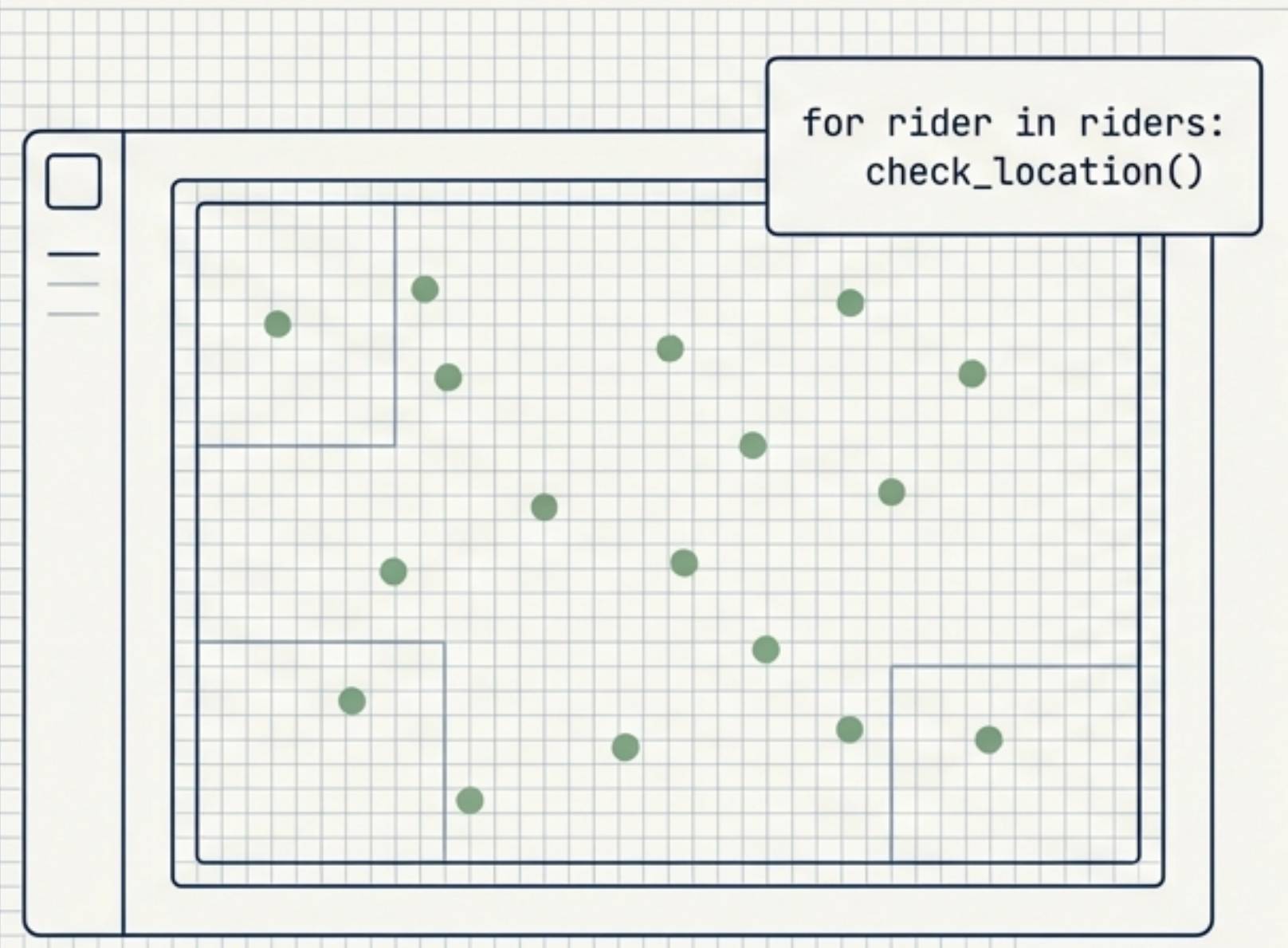


JetBrains Mono

The bridge between "It works on my machine" and "It works for a million users".

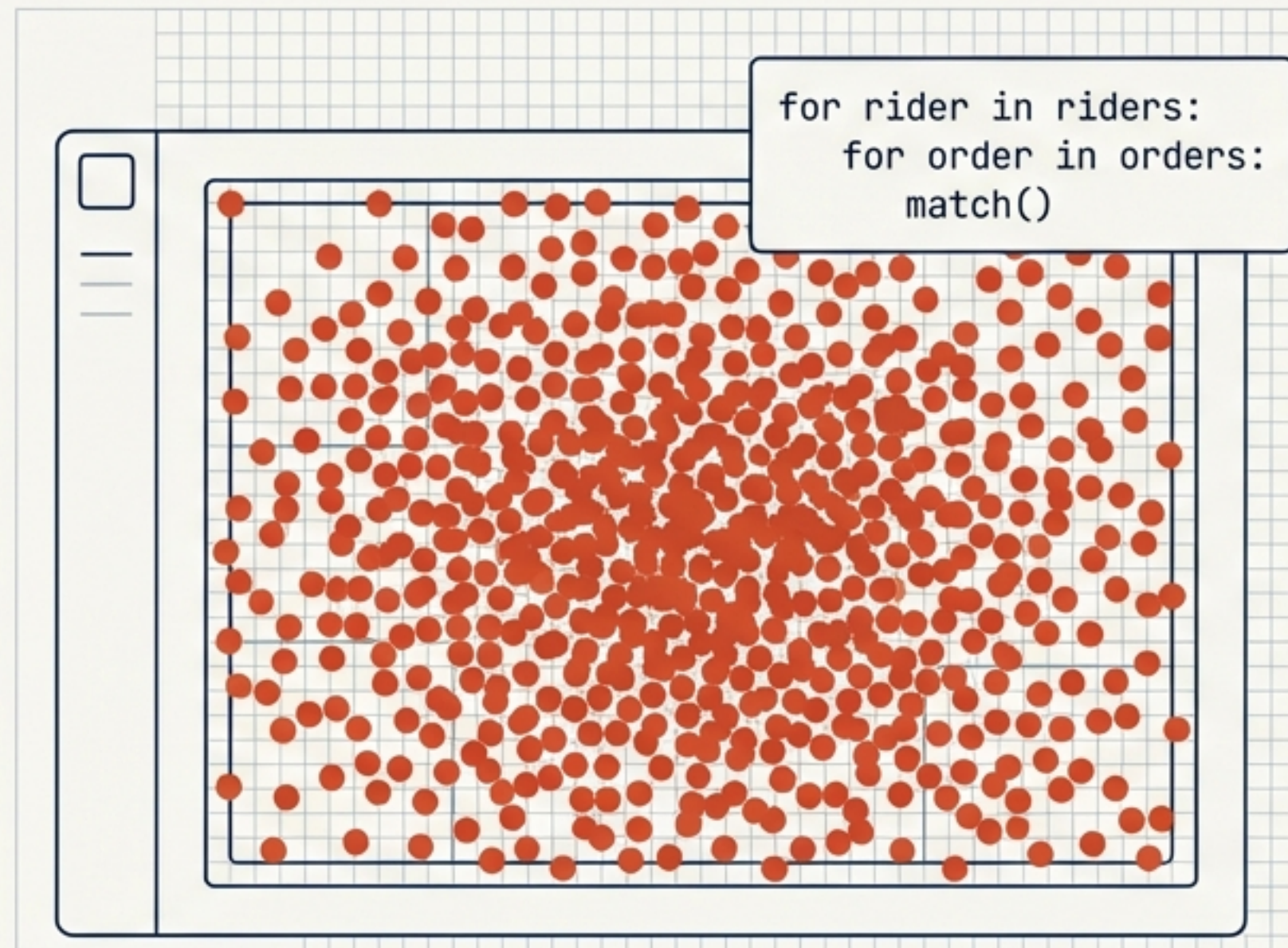
The Startup Trap: When “Working” Code Fails

```
for rider in riders:  
    check_location()
```



FAST (100 Operations)

```
for rider in riders:  
    for order in orders:  
        match()
```



CRASH (1,000,000 Operations)

Nested Loops ($O(N^2)$) work for prototypes but fail at scale.

Algorithms in Your Pocket

Google Maps

Intersection A

Heavy Edge Weight
(Traffic Density)

Intersection B

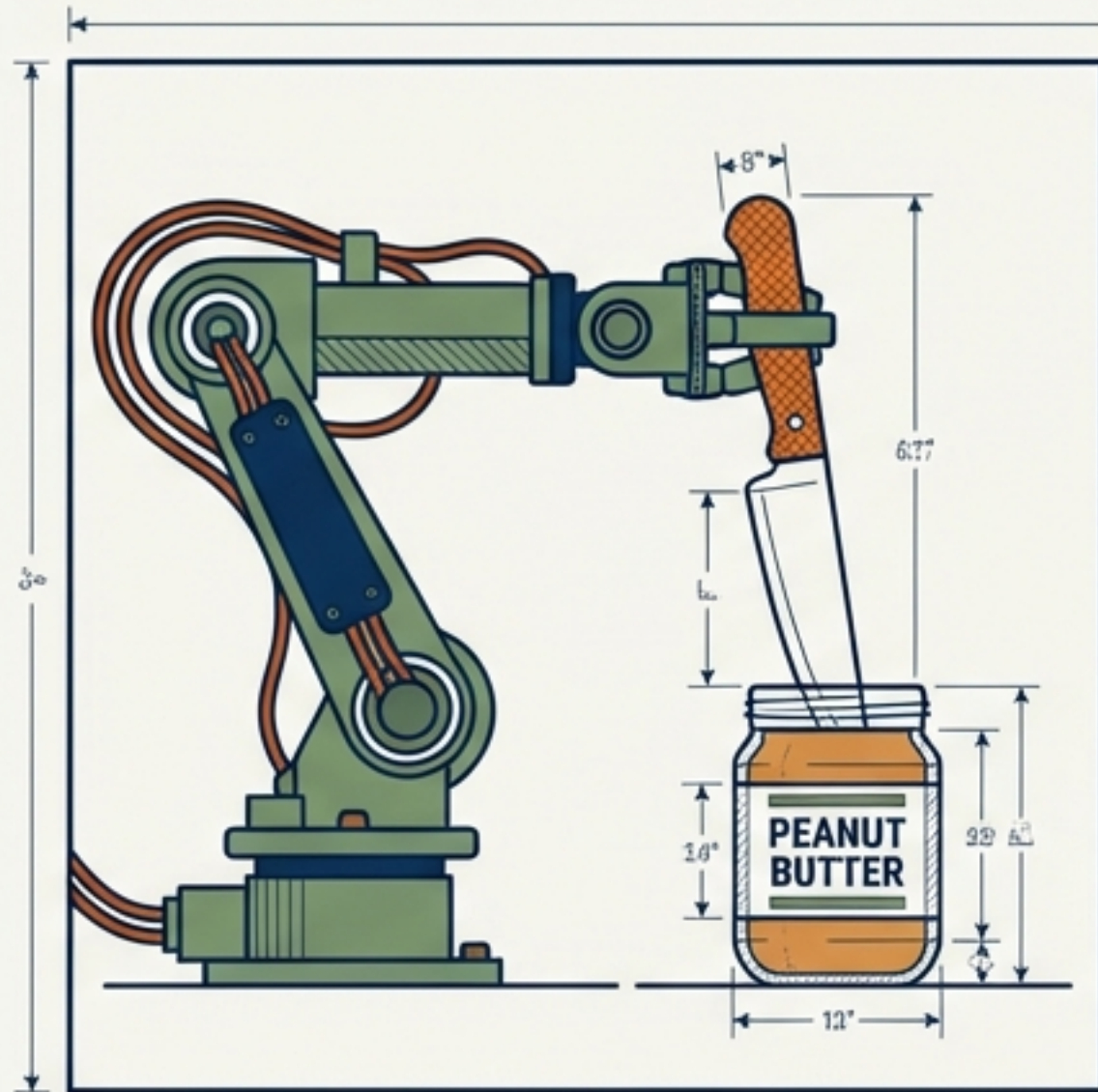
Graph Theory: Navigating nodes based on real-time weights.

Spotify

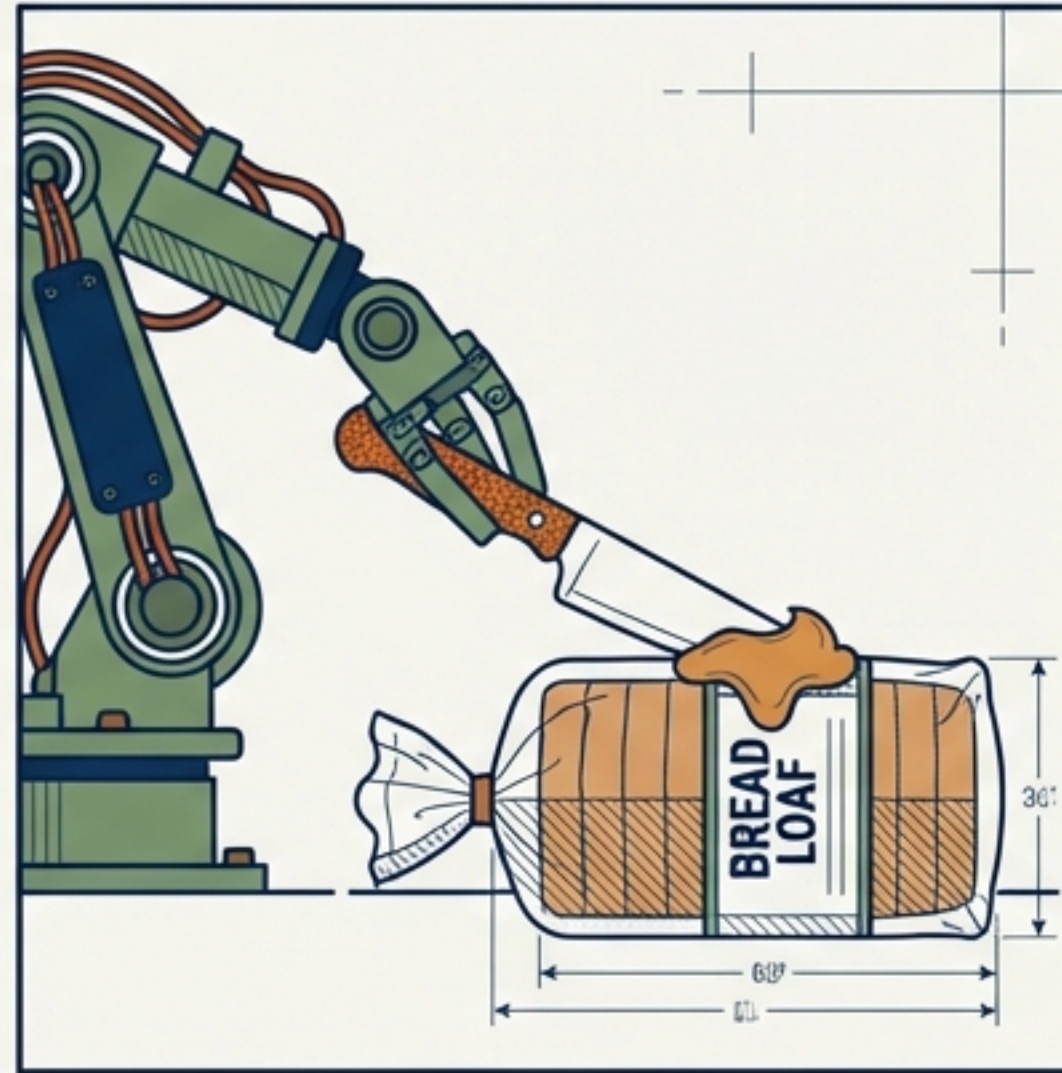
Vector Matching

Linear Algebra: Matching audio vectors to user taste profiles.

The Precision of Instructions



Error: Handle undefined.

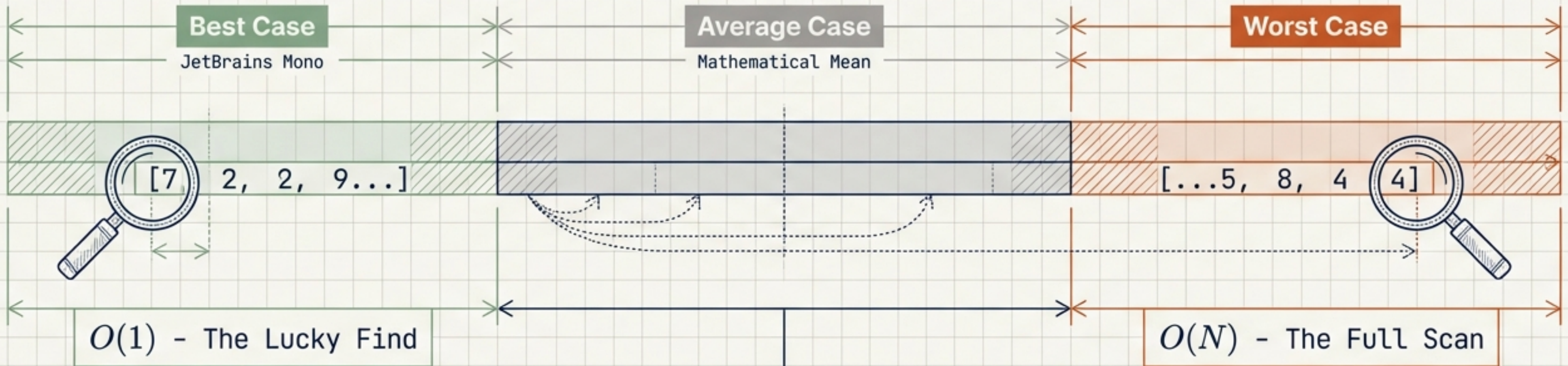


Error: Object not accessible.

Algorithms require
**Algorithmic
Formulation.**

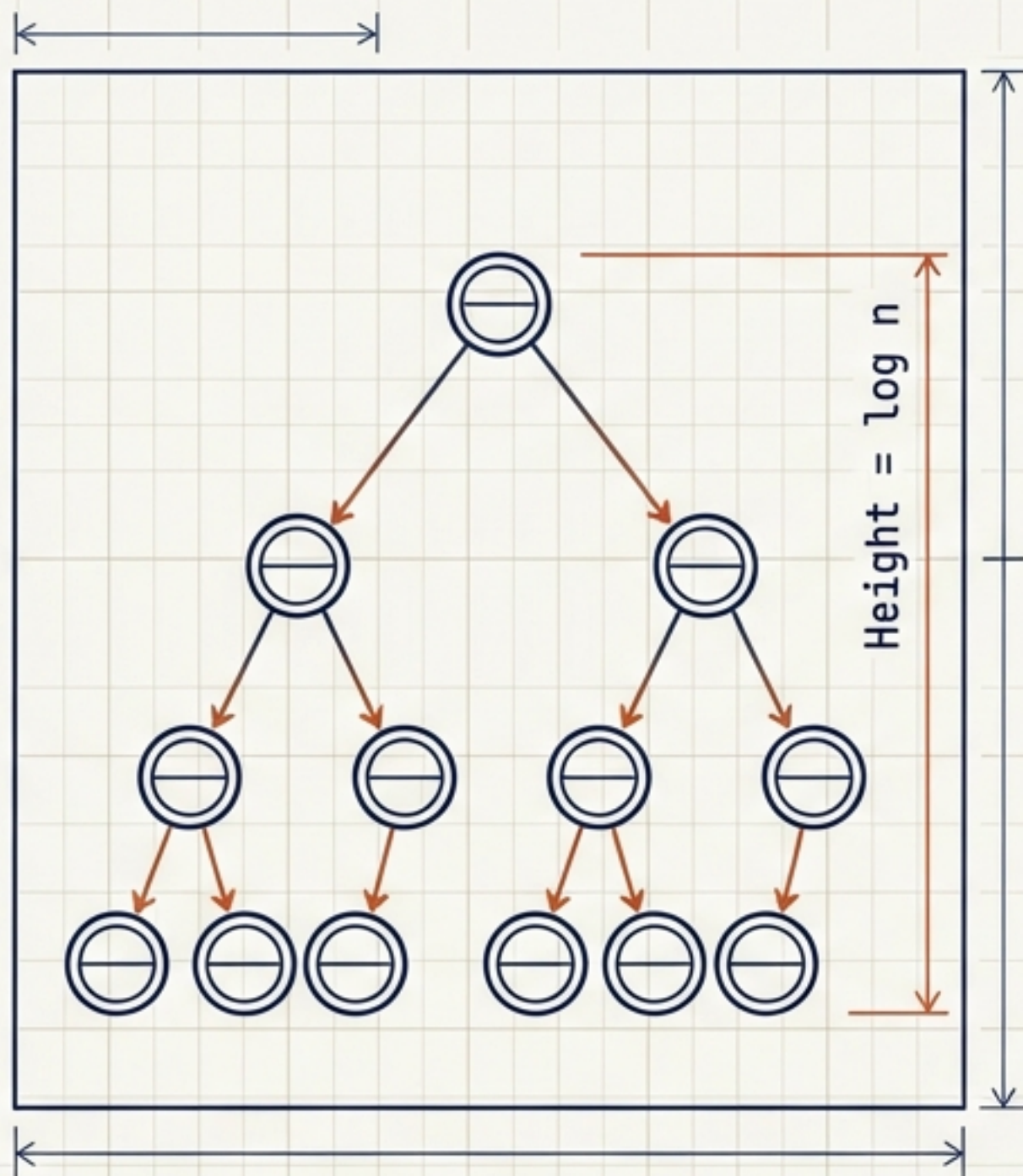
Unambiguous,
step-by-step
instructions that
account for every
state. Computers do
not have intuition;
they only have
instructions.

Defining the Boundaries of Performance



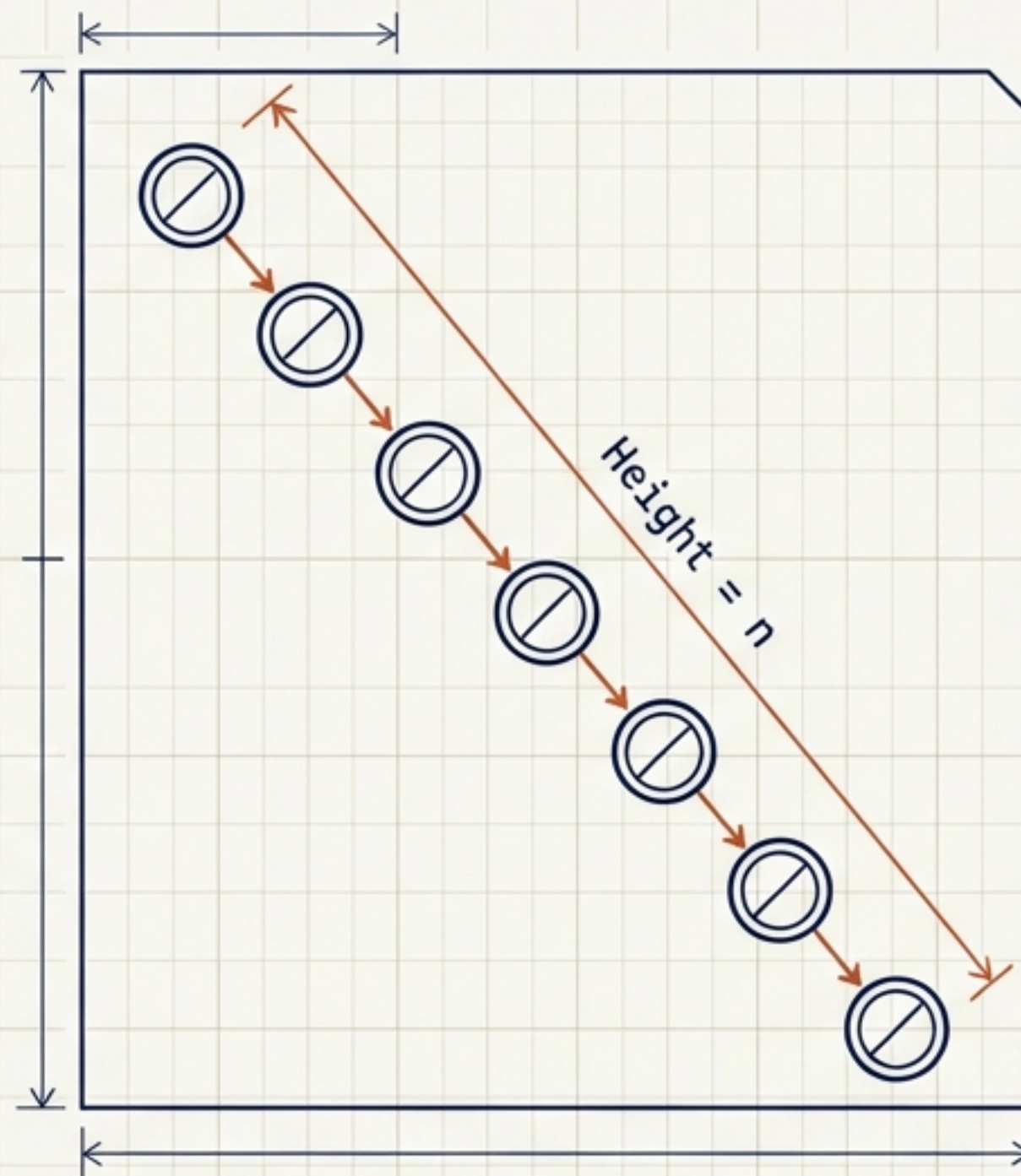
We engineer for the **Worst Case** to prevent the **Startup Crash**.

Structure Dictates Speed: The Binary Search Tree



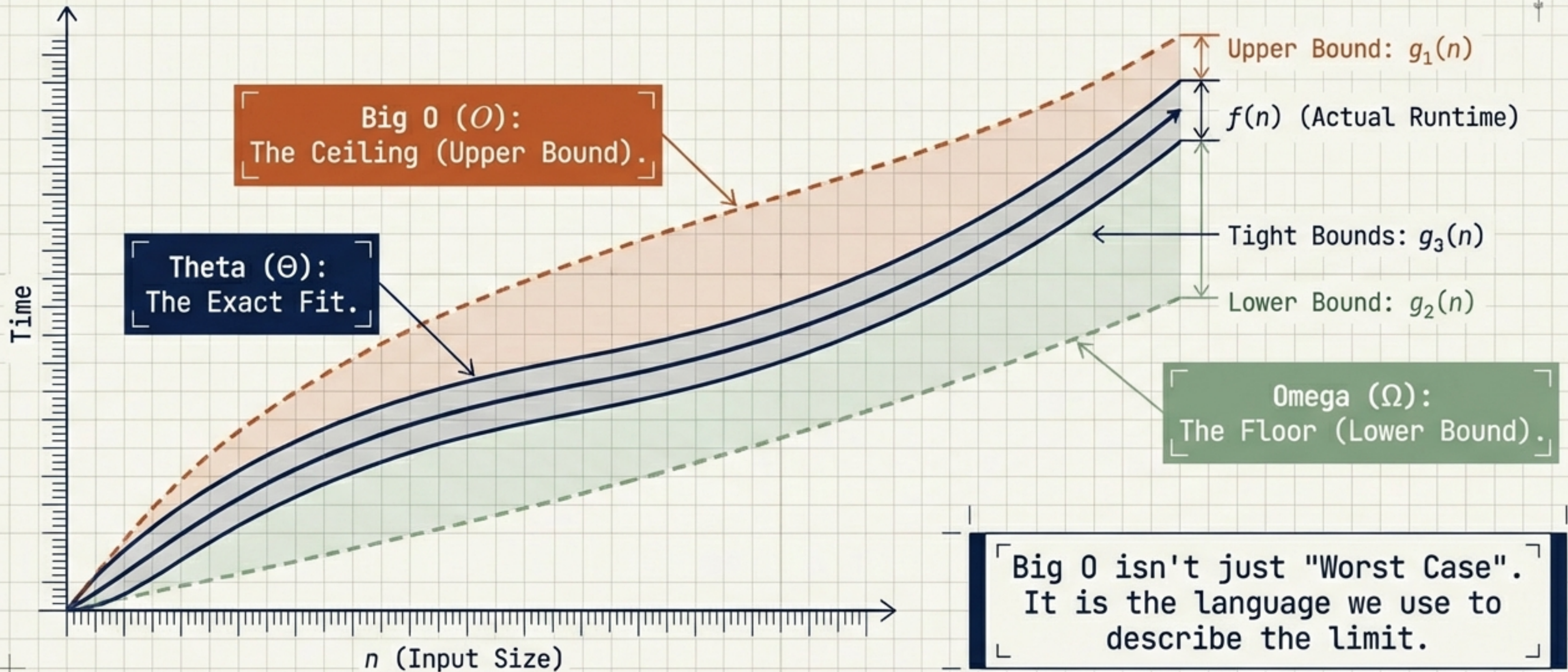
Efficient Search

The speed of the algorithm depends on the shape of the data.



Slow Linear Search

Asymptotic Notations De-mystified



Memory Wars: Arrays vs. Linked Lists

Arrays (The Theater)

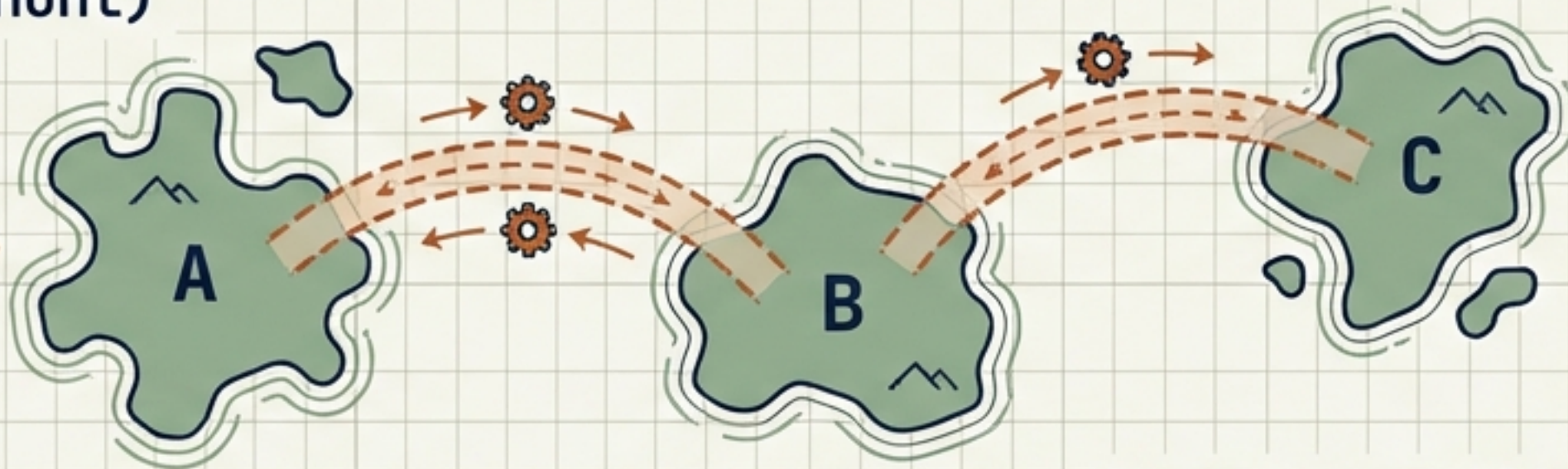


Contiguous Memory:
Fixed Size.
Fast Access, Slow
Expansion.

No space to sit in this specific row.

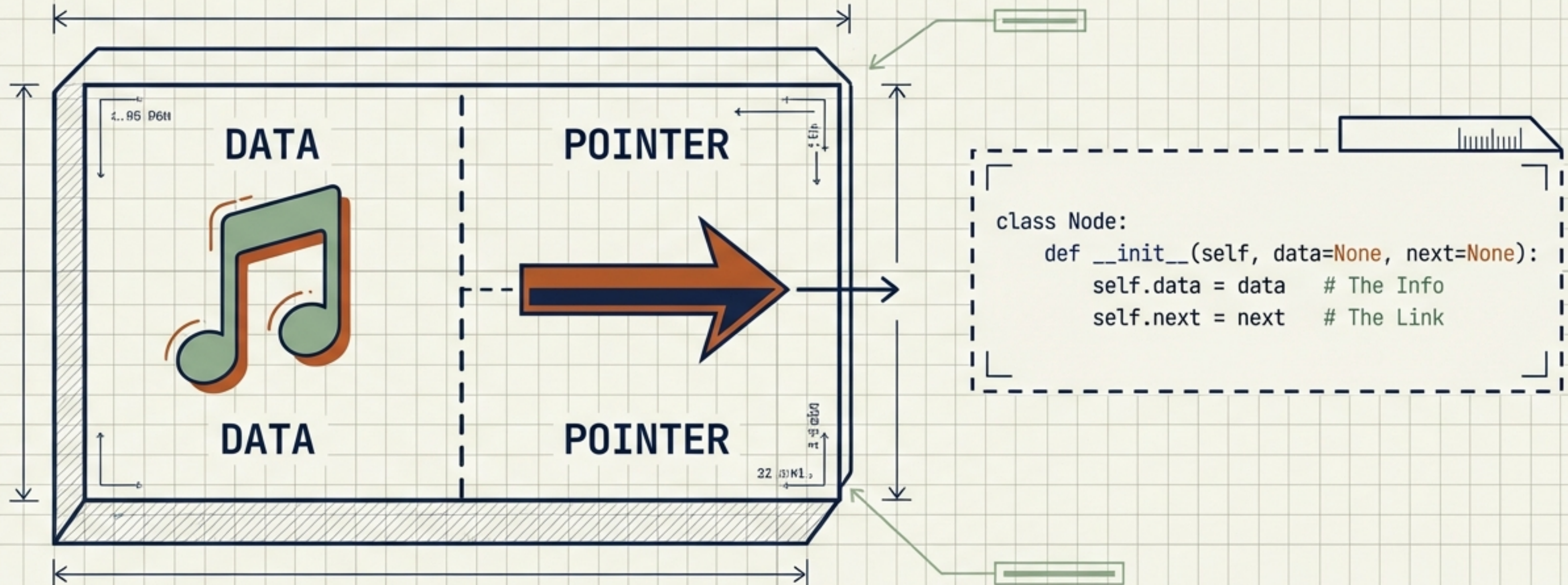
Linked Lists (The Treasure Hunt)

Non-Contiguous Memory:
Dynamic Size.
Flexible Expansion.



Linked but separated islands.

Anatomy of a Node



The atomic building block of a Linked List.

Traversal Logic: The Boss & The Assistant



HEAD (The Boss).
Cannot move.



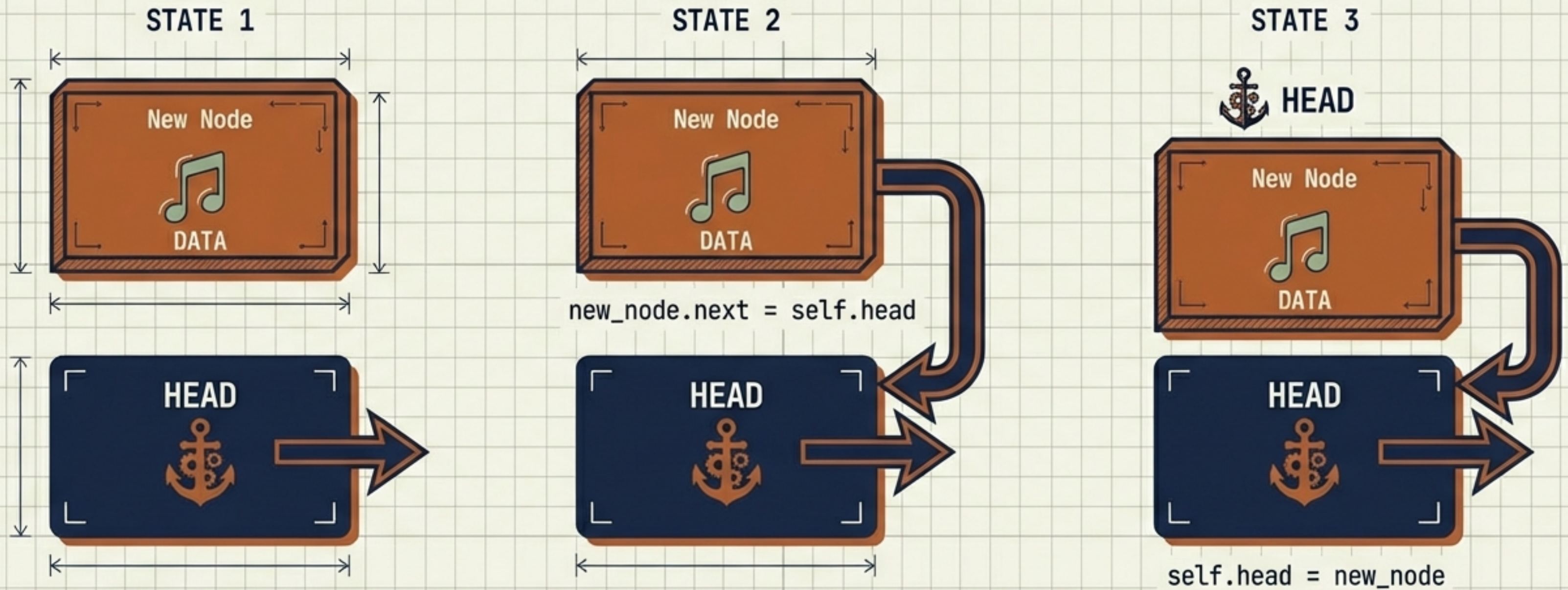
TEMP (The Assistant).
Moves node to node.

PYTHON TRAVERSAL

```
temp = self.head  
while temp is not None:  
    print(temp.data)  
    temp = temp.data  
    temp = temp.next # Move forward
```

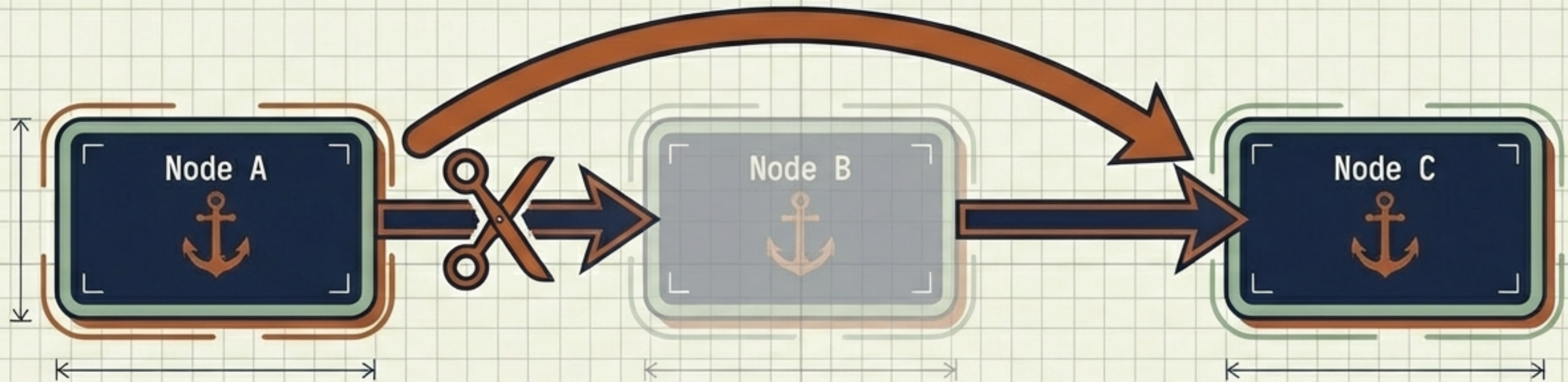


Surgical Precision: Insertion



Unlike arrays, no elements need to shift. We simply rewire the pointers.

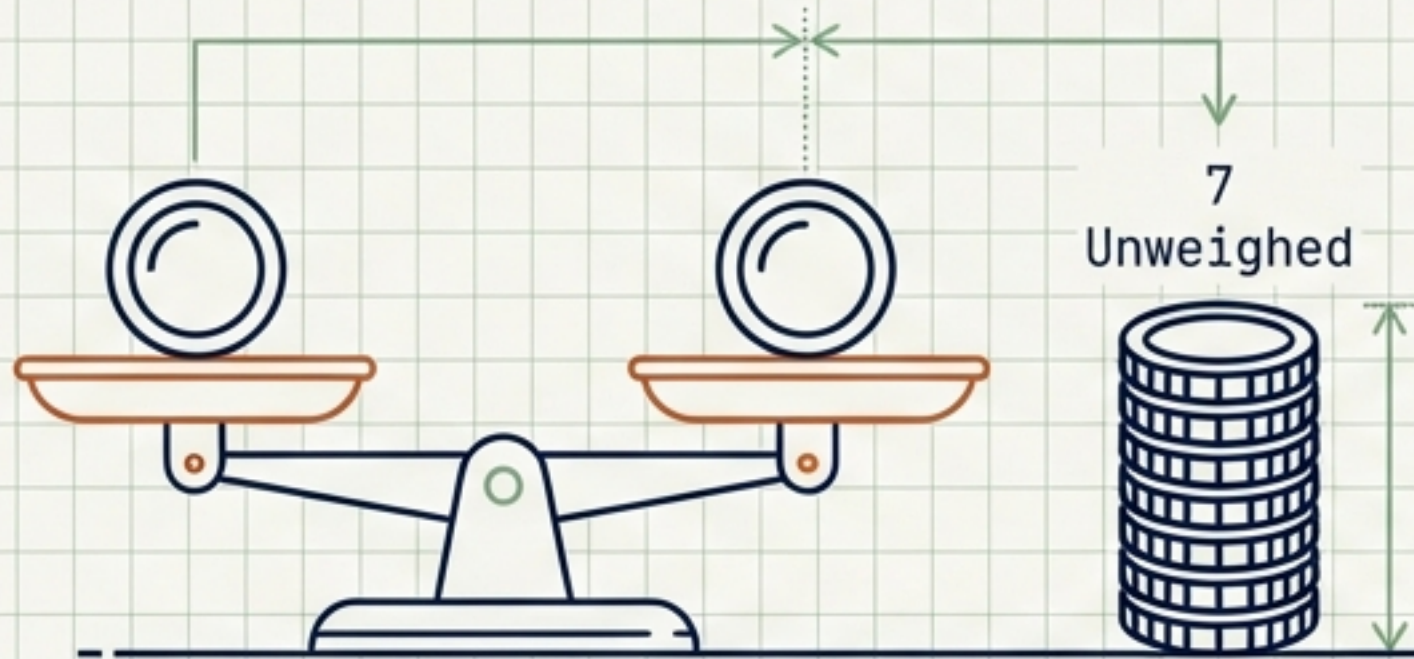
Breaking the Chain: Deletion



```
# Skipping Node B  
previous_node.next = current_node.next
```

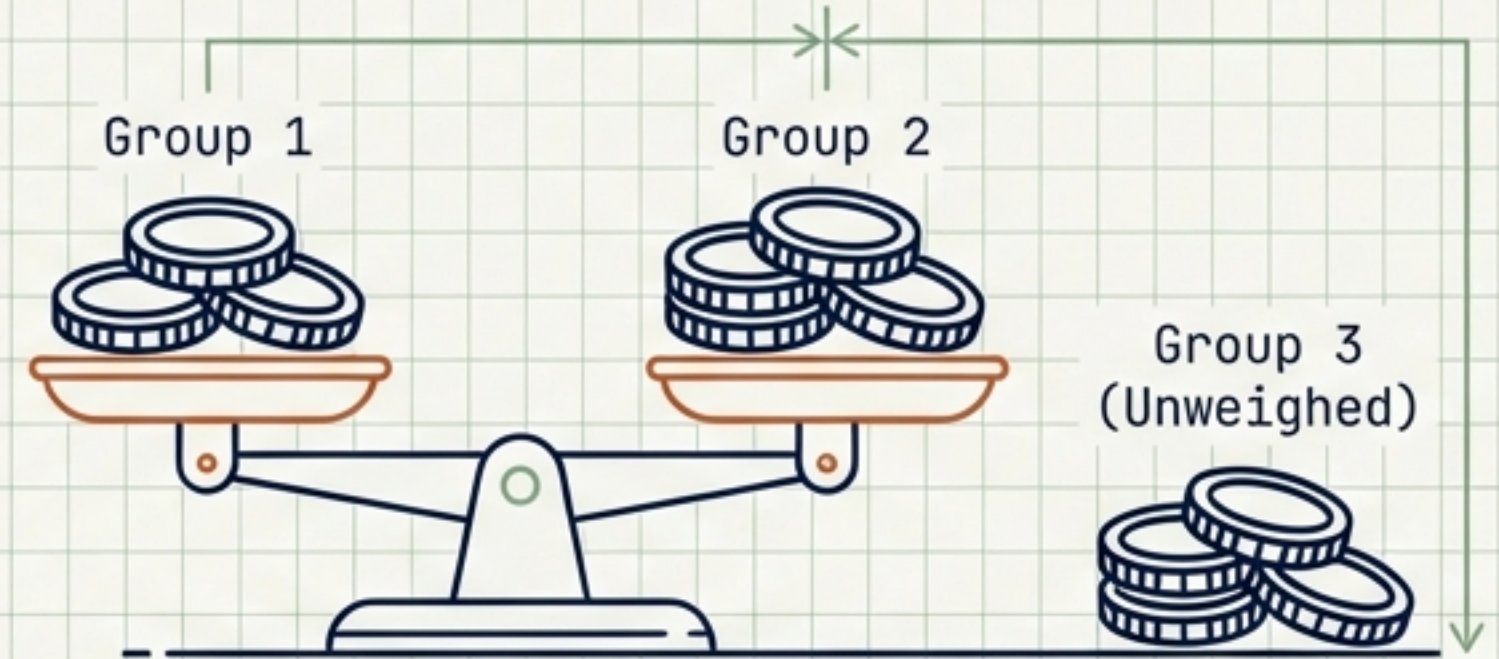

The Logic Puzzle: 9 Coins

Brute Force (Linear)



1 vs 1. Requires up to 4 weighings. $O(N)$.

Divide & Conquer (Algorithmic)



Group weighing. Solved in exactly 2 weighings. $O(\log n)$.

📌 Insight: Smart algorithms reduce the search space exponentially.

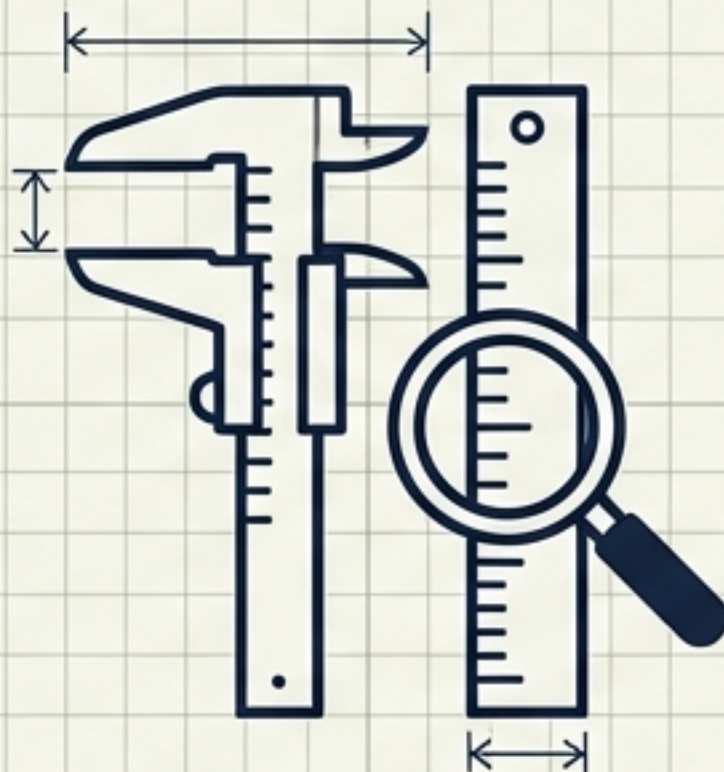
From Scale to Syntax

The Problem



Real-world scale breaks naive code. We need efficiency.

The Measure



Big O notation predicts the breaking point before it happens.

The Solution



Structures like Linked Lists offer dynamic, surgical memory management.

⚓ You don't need to know how water is made to drink it, but you do need to know how to build the pipes if you want to supply the city.